IKT215 Assignment 2:
Parametric and non-parametric classification

Assignments equally contribute to 50% of your final grade. ***Please make sure to read the additional instructions provided towards the end.***

Task 1: Quadratic Discriminant Analysis (45%)

Description:

Using the Wine Quality dataset (available here), implement a parametric plug-in classifier using Quadratic Discriminant Analysis, from scratch. The goal is to classify the wine into good (quality≥5) and bad (quality<5):

1. Select appropriate features (Justify your choice in the report)
2. Split the dataset into training and testing sets.
3. Estimate Class Parameters (Must be included in the report):
    a. Calculate priors
    b. Calculate the MLE of the mean vectors
    c. Calculate the MLE of the covariance matrices
4. Build the classifier:
    a. Compute discriminant function
    b. Assign the class with the highest discriminant score.
5. Train the classifier on the train set and test it on the test set.
6. Report performance of the classifier on the test data with appropriate metrics and visualization.

Task 2: *k*-Nearest Neighbors Classifier (45%)

Description:

Using the Wine Quality dataset, implement a k-nearest neighbors classifier, from scratch. The goal is to classify the wine into good (quality≥5) and bad (quality<5):

1. Use the same features as in Task 1
2. Split the dataset into training and testing sets.
3. Standardize/normalize features, if required.
4. Build the classifier using appropriate distance metric.
5. Train the classifier on the train set using an appropriate value of k and test it on the test set (Justify the k-value in report).
6. Report performance of the classifier on the test data with appropriate metrics and visualization.
7. Compare the performance of the k-nearest neighbors classifier with the parametric plug-in classifier in Task 1.

Additional Instructions

1. Prepare a report (4 pages maximum, excluding front matter).
2. Compile the report (PDF) and your code as a single zip file and name your file "`lastname_firstname.zip`" before uploading.
3. The report should include everything required for the completion of these two tasks.
4. Make sure to report any hyperparameter choices you made.
5. Only include the relevant visualizations in the report.
6. The visualizations in the report should be legible.

NB: Submit your code as a jupyter notebook file ("`*.ipynb`") with outputs saved. Otherwise, scripts could run into failures due to dependencies on execution. Instructions for running a jupyter notebook are provided on Canvas.

Rubric

Task	**Score (%)**
Task 1	
Feature selection and justification	10
Parameters estimation	15
Classifier definition	5
Performance evaluation, visualization, and discussion	15
Task 2	
Choice of distance metric and justification	10
Selection of k and justification	15
Performance evaluation, visualization, and discussion	15
QDA and k-NN comparison	5
Report quality and structure	10
Total	**100**

Use of Generative AI:

While the use of generative AI tools is not prohibited, it is strongly discouraged for this assignment. All reports must include **a dedicated disclaimer page** that honestly describes which AI tools (if any) were used and how they were applied, in accordance with UiA's general guidelines. Remember that **submitting AI-generated text without proper citation constitutes plagiarism and will be treated as such.**

Late Submission Policy

Late submissions will incur a **5% penalty per calendar day**. Submissions will not be accepted more than six days after the deadline. Plan accordingly and contact the instructor in advance if you anticipate difficulties meeting the deadline.